# Assignment 4: Sequential Modeling with Recurrent Networks

Course: Large Learning Models

Assignment Report

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# Overview

This report documents two sequential-classification experiments implemented in PyTorch. Task 1 converts Kannada handwritten digit images into sequences of image rows. Task 2 classifies spoken Arabic digits from variable-length sequences of 13-dimensional MFCCs. The notebooks compare RNN, LSTM, and GRU models across 36 configurations for Task 1 and 18 for Task 2.

Both notebooks use Adam and cross-entropy loss and stop when the absolute difference between consecutive average training losses is below $10^{-4}$. Neither notebook uses validation accuracy for model selection. Both rank configurations by test accuracy. Task 2's best model reaches 100.00% training accuracy but only 12.18% test accuracy; this result is reported as observed and discussed cautiously.

# TASK 1: Kannada Digit Classification with RNN, LSTM and GRU

## 1. Objectives

Classify ten Kannada digit classes by treating each 28 × 28 grayscale image as 28 time steps, each with 28 pixel features. Compare cell types, hidden sizes, and layer counts using training/test accuracy, loss curves, confusion matrix, and sample visualizations.

## 2. Dataset and Preprocessing

The notebook reads train.csv and reports 42,000 training, 9,000 validation, and 9,000 test samples. Splits use random_state=42 but are not stratified. Pixel values are reshaped to 28 × 28 and divided by 255, giving inputs of shape (batch, 28, 28). The validation loader is created but not used in training or model selection.

| Subset | Samples | Role |
|---|---|---|
| Training | 42,000 | Gradient updates |
| Validation | 9,000 | Created but not used |
| Test | 9,000 | Reported test accuracy |

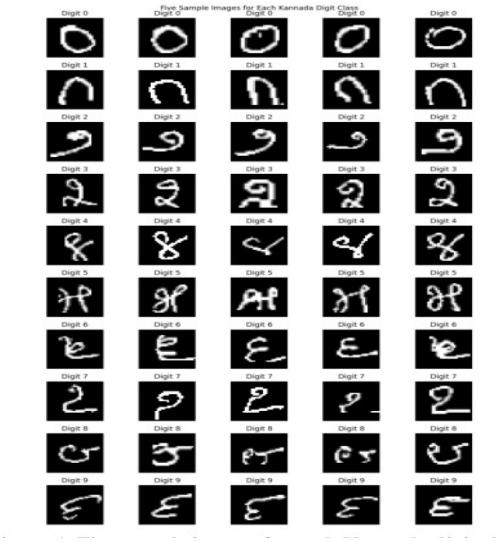


*Figure 1. Five sample images for each Kannada digit class.*

# 3. Methodology

## 3.1 Recurrent Network Design

The model uses an RNN, LSTM, or GRU stack. The final timestep output is passed to a linear layer with ten outputs.

## 3.2 Loss Function and Optimizer

CrossEntropyLoss is optimized with Adam at learning rate 0.001 and batch size 128.

## 3.3 Hyperparameter Search

The search includes 3 cell types × 4 hidden sizes (32, 64, 128, 256) × 3 layer counts (1, 2, 3), for 36 configurations. The stopping rule is $|\bar{L}(k) - \bar{L}(k-1)| < 10^{-4}$. The loop has no maximum epoch limit.

| Parameter | Setting |
|---|---|
| Cell type | RNN, LSTM, GRU |
| Hidden size | 32, 64, 128, 256 |
| Layers | 1, 2, 3 |
| Optimizer | Adam |
| Learning rate | 0.001 |
| Batch size | 128 |
| Loss | Cross-entropy |
| Stopping | Consecutive average-loss difference < $10^{-4}$ |

# 4. Experimental Results and Analysis

The table is sorted by test accuracy, matching the notebook. The best row is GRU with 256 units and 3 layers, reporting 99.95% training and 99.39% test accuracy. Since test accuracy is used for selection, the reported test score is not an independent post-selection estimate.

| Cell | Units | Layers | Epochs | Train % | Test % |
|---|---|---|---|---|---|
| GRU | 256 | 3 | 30 | 99.95 | 99.39 |
| LSTM | 32 | 3 | 95 | 100.00 | 99.38 |
| LSTM | 64 | 3 | 79 | 100.00 | 99.34 |
| GRU | 128 | 1 | 61 | 100.00 | 99.32 |
| GRU | 128 | 2 | 38 | 99.99 | 99.29 |
| LSTM | 128 | 3 | 31 | 99.93 | 99.27 |
| LSTM | 128 | 2 | 26 | 99.80 | 99.26 |
| GRU | 64 | 3 | 31 | 99.92 | 99.18 |
| GRU | 128 | 3 | 19 | 99.82 | 99.10 |
| GRU | 256 | 1 | 13 | 99.65 | 99.09 |
| LSTM | 256 | 1 | 63 | 99.91 | 99.04 |
| GRU | 256 | 2 | 41 | 99.90 | 99.04 |
| GRU | 64 | 2 | 37 | 99.92 | 99.03 |
| LSTM | 128 | 1 | 18 | 99.48 | 98.99 |
| LSTM | 64 | 2 | 35 | 99.75 | 98.97 |
| GRU | 64 | 1 | 30 | 99.74 | 98.96 |
| GRU | 32 | 3 | 68 | 99.86 | 98.96 |
| GRU | 32 | 2 | 47 | 99.90 | 98.94 |
| RNN | 128 | 1 | 130 | 99.81 | 98.86 |
| LSTM | 256 | 3 | 11 | 99.36 | 98.82 |
| LSTM | 256 | 2 | 12 | 99.40 | 98.81 |
| LSTM | 64 | 1 | 59 | 99.77 | 98.81 |
| LSTM | 32 | 2 | 68 | 99.95 | 98.79 |
| GRU | 32 | 1 | 85 | 99.92 | 98.59 |
| RNN | 128 | 3 | 53 | 99.04 | 98.47 |
| RNN | 64 | 3 | 144 | 99.16 | 98.44 |
| RNN | 32 | 3 | 101 | 99.38 | 98.43 |
| LSTM | 32 | 1 | 34 | 99.11 | 98.31 |
| RNN | 128 | 2 | 21 | 98.85 | 98.27 |
| RNN | 32 | 2 | 46 | 98.72 | 98.04 |
| RNN | 256 | 1 | 140 | 98.36 | 97.84 |
| RNN | 256 | 2 | 49 | 98.25 | 97.82 |
| RNN | 64 | 1 | 60 | 98.70 | 97.72 |
| RNN | 64 | 2 | 16 | 98.05 | 97.68 |

| | | | | | |
|---|---|---|---|---|---|
| RNN | 256 | 3 | 106 | 96.76 | 96.39 |
| RNN | 32 | 1 | 18 | 95.34 | 94.40 |

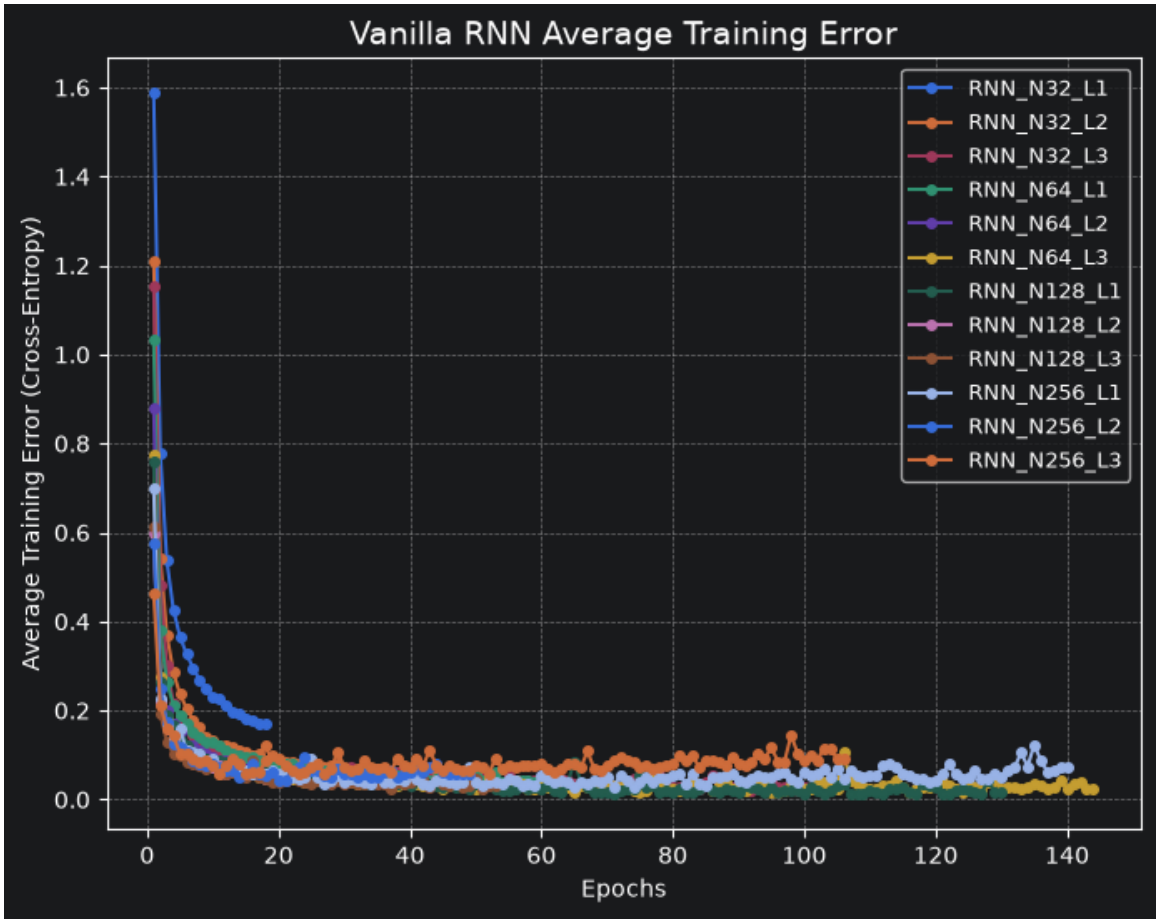



*Figure 2. Average training error for RNN configurations.*

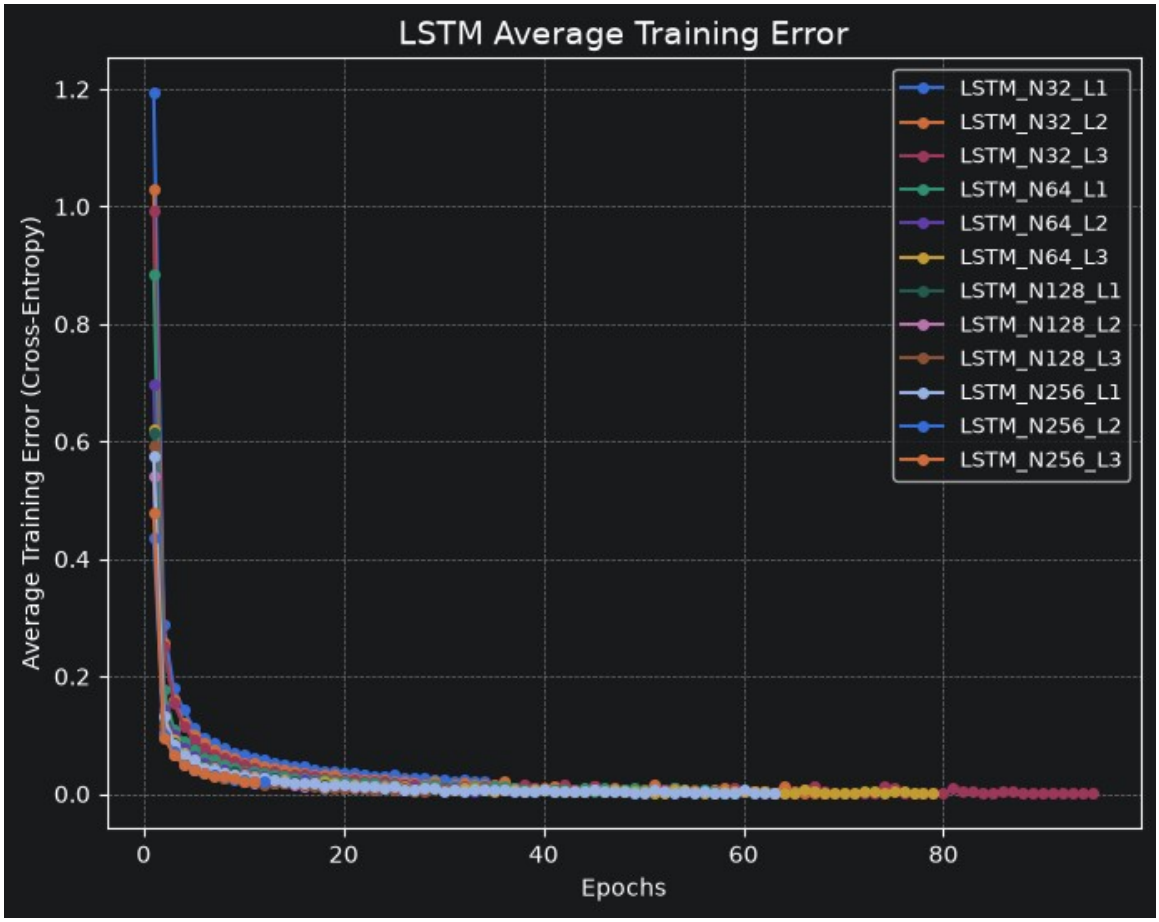



*Figure 3. Average training error for LSTM configurations.*

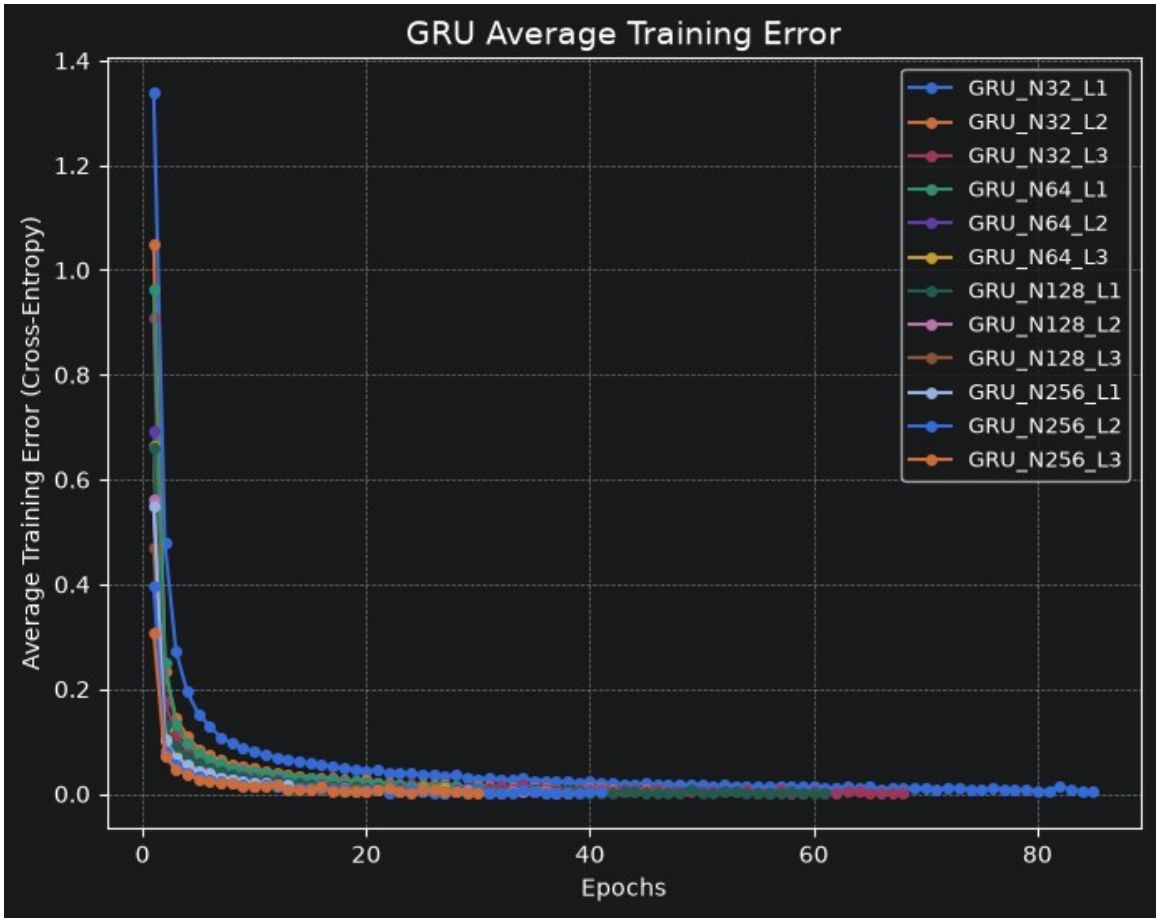



*Figure 4. Average training error for GRU configurations.*

## 4.1 Best Configuration and Confusion Matrix

The selected configuration is GRU_N256_L3 with 99.95% training accuracy and 99.39% test accuracy. The notebook does not compute validation accuracy or a per-class classification report.

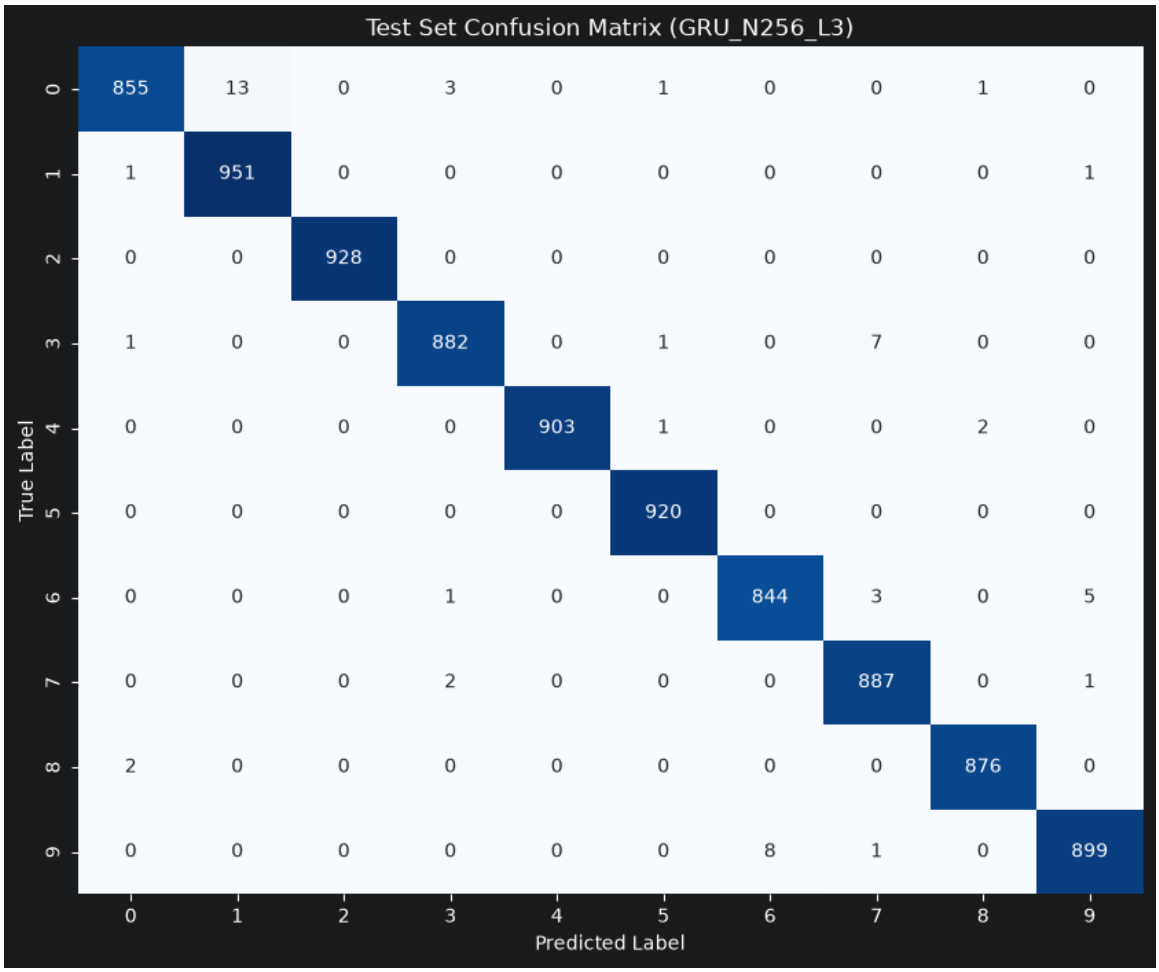



*Figure 5. Test confusion matrix for GRU_N256_L3.*

*Figure 6. Qualitative Kannada digit samples.*

## 5. Task 1 Conclusion

The best recorded Task 1 test accuracy is 99.39%. The validation split is not used, and models are selected by test accuracy. Future evaluation should select on validation performance, reserve the test set for final evaluation, set seeds, and add a maximum epoch limit.

# TASK 2: Spoken Arabic Digit Recognition with Recurrent Networks

## 1. Objectives

Classify spoken Arabic digits from variable-length sequences of 13 MFCC coefficients and compare RNN, LSTM, and GRU models across hidden sizes and layer counts.

## 2. Dataset and Preprocessing

The notebook loads 6,600 training utterances and 2,200 test utterances from text files. Blank lines delimit utterances. Each frame is parsed as 13 numerical features. Labels are assigned by (index // 10) % 10, which assumes the file order matches groups of ten repetitions per digit. Variable-length sequences are padded per batch, sorted by length, and passed through pack_padded_sequence. No validation split or MFCC standardization is implemented.

| Subset | Utterances | Role |
|---|---|---|
| Training | 6,600 | Gradient updates |
| Test | 2,200 | Reported test accuracy |

## 3. Methodology

The recurrent classifier has 13 input features, one or two recurrent layers, hidden sizes 32, 64, or 128, and a ten-class linear output. The 18 models use CrossEntropyLoss, Adam at 0.001, batch size 64, and the same $10^{-4}$ loss-change stopping rule. There is no maximum epoch limit.

| Parameter | Setting |
|---|---|
| Cell types | RNN, LSTM, GRU |
| Hidden size | 32, 64, 128 |
| Layers | 1, 2 |
| Optimizer | Adam, learning rate 0.001 |
| Batch size | 64 |
| Stopping | Consecutive average-loss difference < $10^{-4}$ |

## 4. Experimental Results and Analysis

The best notebook result is LSTM_N64_L2 with 100.00% training accuracy and 12.18% test accuracy. Results remain near the 10% chance baseline for ten classes, indicating poor generalization in the current pipeline.

| Cell | Units | Layers | Epochs | Train % | Test % |
|---|---|---|---|---|---|
| LSTM | 64 | 2 | 103 | 100.00 | 12.18 |
| LSTM | 128 | 2 | 62 | 99.98 | 11.00 |
| GRU | 64 | 1 | 165 | 100.00 | 11.00 |
| RNN | 64 | 2 | 925 | 99.42 | 10.91 |
| LSTM | 64 | 1 | 194 | 100.00 | 10.91 |
| GRU | 32 | 2 | 174 | 99.55 | 10.59 |
| RNN | 32 | 2 | 525 | 65.74 | 10.55 |
| LSTM | 128 | 1 | 86 | 100.00 | 10.41 |
| RNN | 128 | 1 | 776 | 56.06 | 10.36 |
| RNN | 64 | 1 | 71 | 56.35 | 10.36 |
| GRU | 128 | 2 | 36 | 100.00 | 10.05 |
| GRU | 32 | 1 | 208 | 70.85 | 9.95 |
| GRU | 64 | 2 | 87 | 100.00 | 9.82 |
| LSTM | 32 | 2 | 342 | 100.00 | 9.68 |
| RNN | 128 | 2 | 96 | 100.00 | 9.50 |
| RNN | 32 | 1 | 338 | 41.97 | 9.41 |
| GRU | 128 | 1 | 51 | 99.98 | 9.14 |
| LSTM | 32 | 1 | 383 | 94.98 | 8.82 |

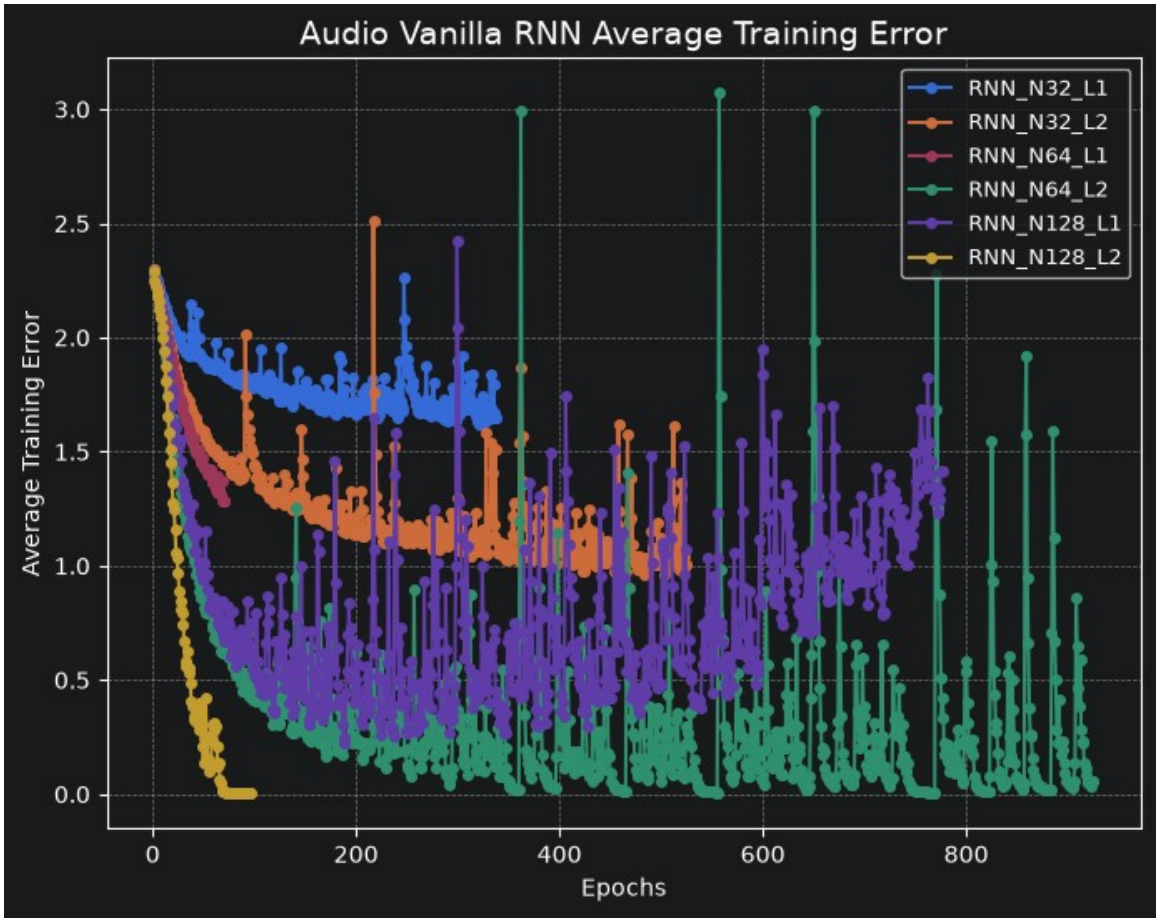



*Figure 7. Average training error for RNN configurations.*

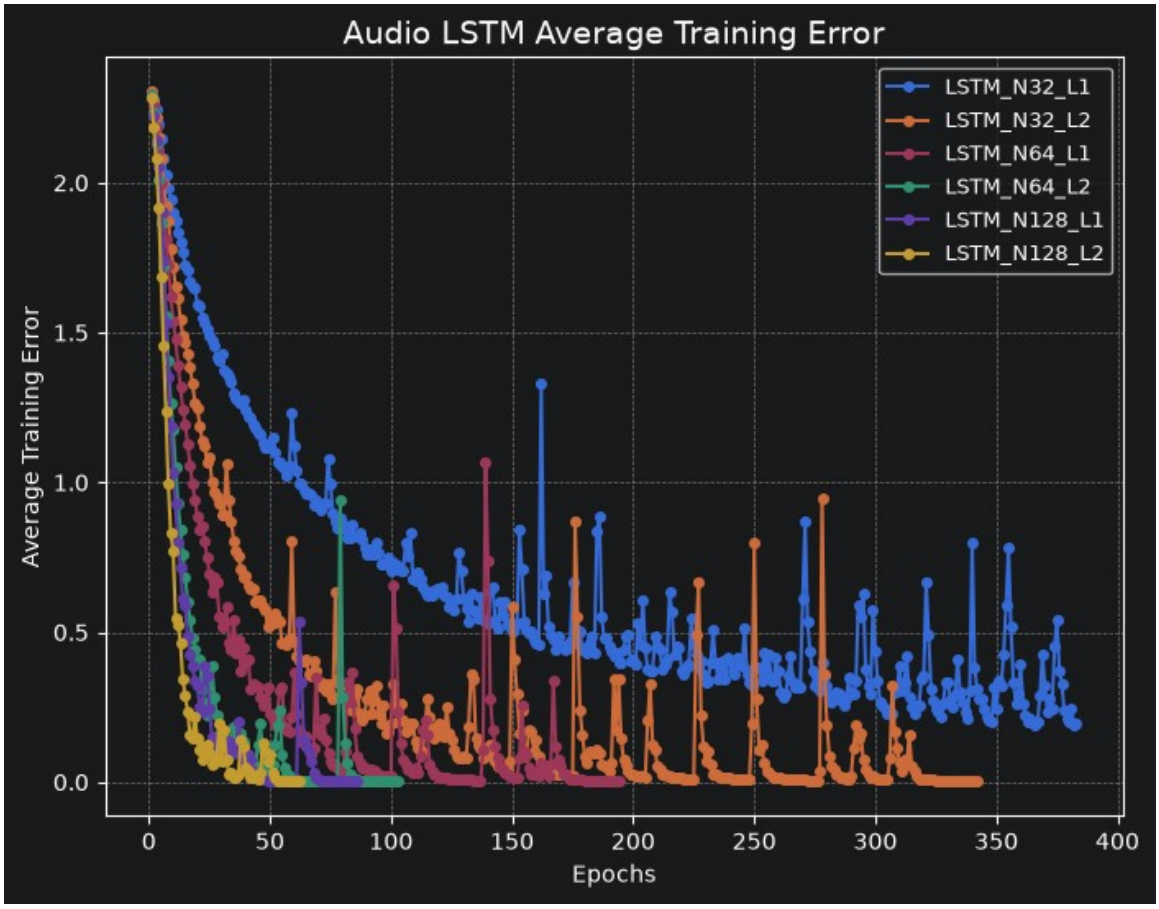



*Figure 8. Average training error for LSTM configurations.*




*Figure 9. Average training error for GRU configurations.*

## 4.1 Best Configuration and Confusion Matrix

LSTM_N64_L2 was selected by sorting the test accuracy table. It ran for 103 epochs and achieved 12.18% test accuracy. The confusion matrix is included below; the notebook does not print per-class precision, recall, or F1-score.

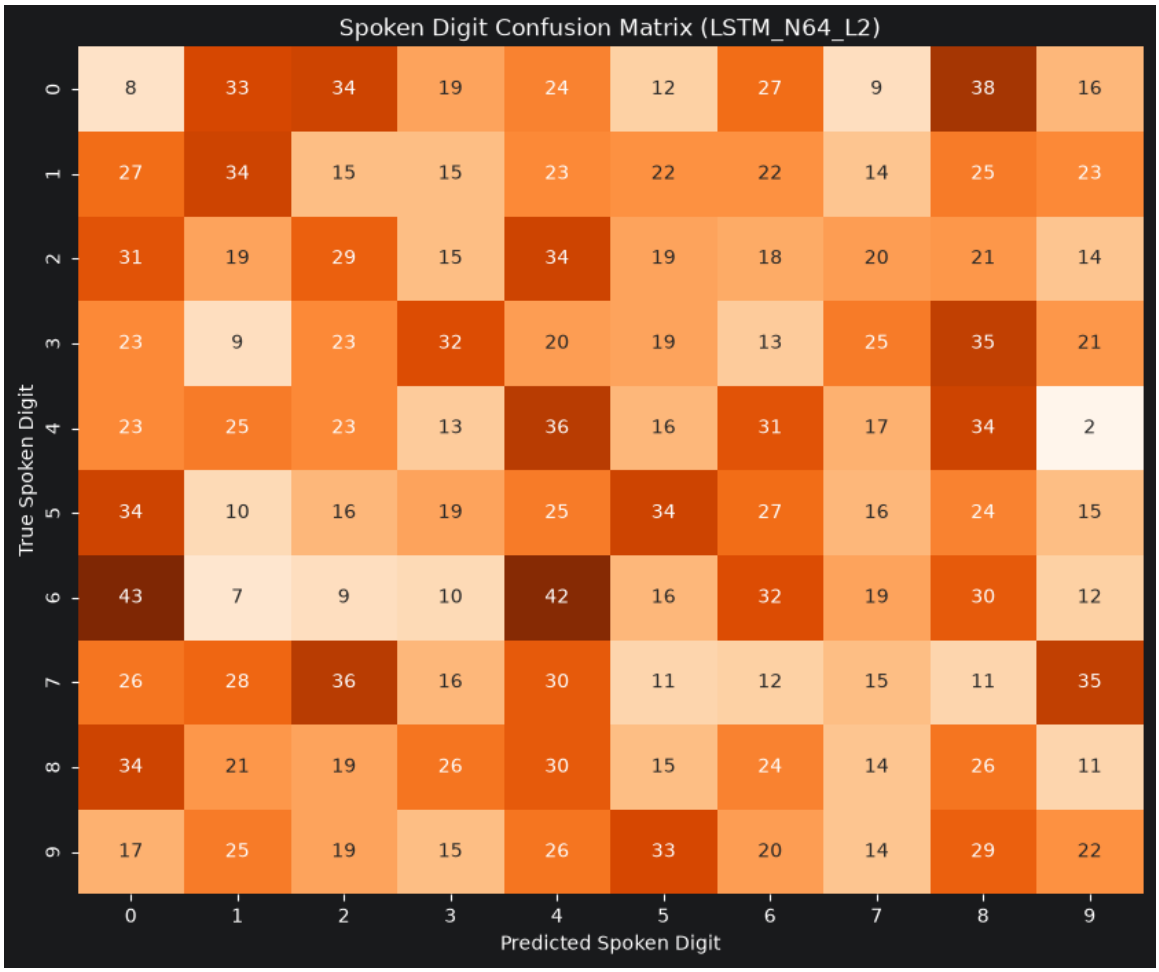



*Figure 10. Test confusion matrix for LSTM_N64_L2.*

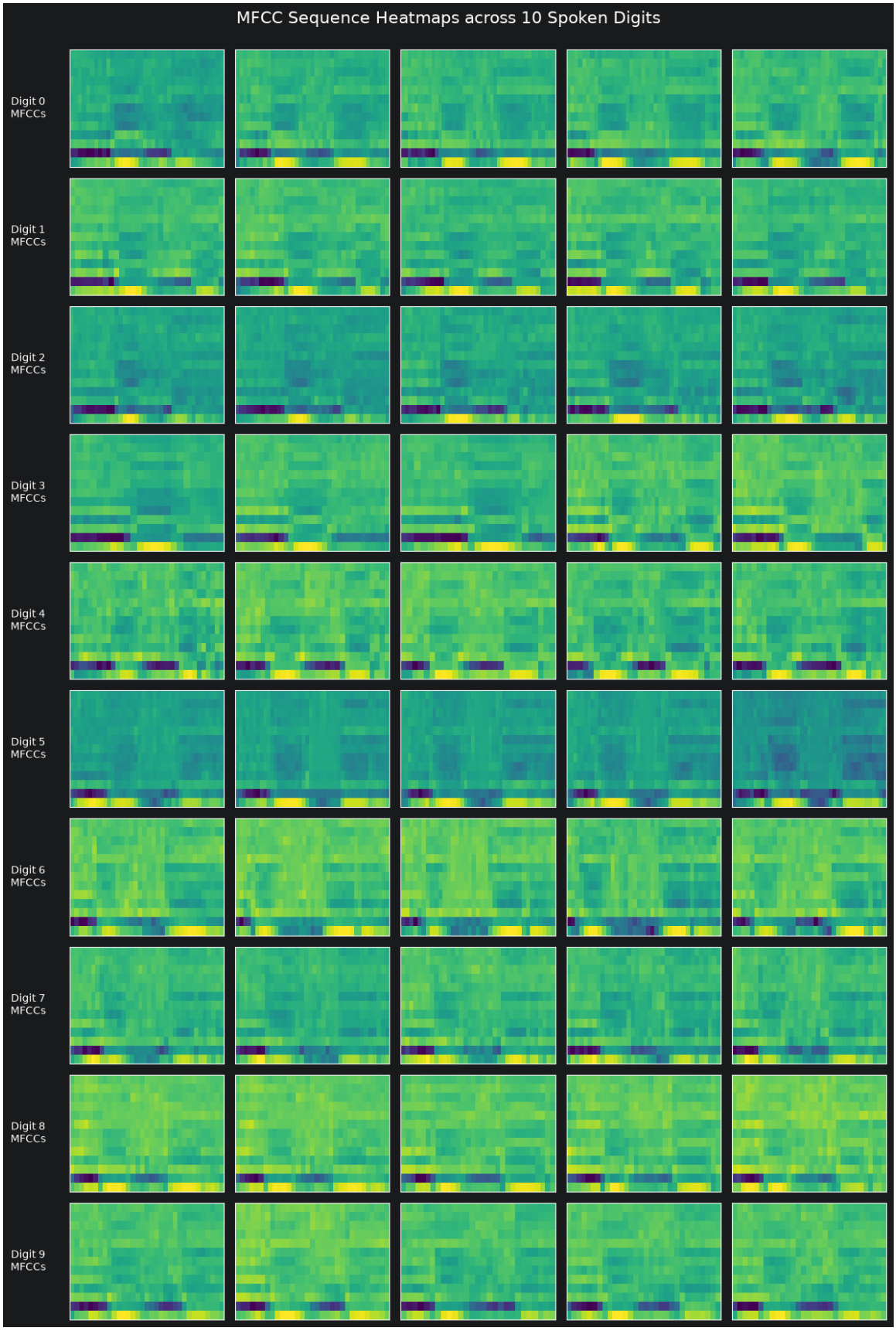



*Figure 11. MFCC heatmaps for sample utterances across the ten digit classes.*

## 5. Limitations and Interpretation

The very low test score and near-perfect training scores indicate a severe train–test gap. The label assignment depends on a fixed utterance ordering; if that assumption does not match the files, labels may be incorrect. This is a possible issue to verify, not a confirmed cause. The code also lacks validation-based selection, MFCC standardization, a random seed, and a maximum epoch limit.

Recommended follow-up checks are to verify label order, inspect labels for sampled utterances, compare class counts, standardize using training-only statistics, create a validation split, and select the model on validation accuracy before a single final test evaluation.

## 6. Task 2 Conclusion

The notebook demonstrates variable-length sequence batching but does not achieve reliable spoken-digit classification in the reported run. The best test accuracy is 12.18%. Label integrity and the evaluation pipeline should be checked before drawing conclusions about the relative merits of RNN, LSTM, and GRU.

# Overall Conclusion

For Task 1, the best recorded test accuracy is 99.39% using a 3-layer GRU with 256 hidden units. For Task 2, the best recorded test accuracy is 12.18% using a 2-layer LSTM with 64 hidden units, despite 100.00% training accuracy. The notebooks include model sweeps, training-loss curves, confusion matrices, and qualitative input visualizations. They do not use validation accuracy for model selection, and Task 2 needs label verification and better preprocessing. These limitations are stated explicitly rather than filled with unsupported claims.